Chenghao (Tommy) Jiang

3D Computer Vision · SLAM · Generative Models

Los Angeles, CA | tommyjiangch@gmail.com | (608) 867-9882 | LinkedIn | JesusmiCaH | jesusmicah.github.io

EDUCATION

University of Wisconsin–Madison — *Sep 2024 — Dec 2025*
MS in Electrical and Computer Engineering — Madison, WI
- GPA: 3.66/4.0

University of Manchester — *Sep 2022 — Dec 2023*
MS in Communication and Signal Processing — Manchester, UK
- GPA: 83.5/100 · Distinction Honor

Changchun University of Science and Technology — *Sep 2018 — Jun 2022*
BEng in Optoelectronic Information Science and Engineering — Changchun, China
- GPA: 3.86/5.00 · Rank: 10/221

EXPERIENCE

Tera AI — *Remote*
3D Vision Researcher — Feb 2026 — Sep 2026
- Contributed to a **GPS-denied SLAM system** for fixed-wing UAVs, with flight validation meeting **position RMSE <30 m** above 150 ft AGL.
- Co-designed a **MASt3R-SLAM-inspired pipeline** combining learned image correspondences with geographic priors. On NVIDIA Jetson Orin NX 16GB, the new deployment achieved **20 FPS**, compared with **<2 FPS** for the previous MASt3R-SLAM deployment.
- Trained the core **flow prediction model** using BlendedMVS, TartanAirV2, Kubric4D, and internal flight data; converted pilot-recorded MCAP logs into BlendedMVS-style training datasets.
- Built a **diagnostic pipeline** after training, linking correspondence predictions, pose estimation, and trajectory APE to analyze system failures and **curate hard cases** for targeted fine-tuning.

Research Intern — Part-time — Aug 2025 — Feb 2026
- Integrated **causal attention** and **KV caching** into a **feed-forward 3D model**, enabling **streaming video** 3D reconstruction and camera localization for the team.
- Independently reimplemented and trained **DINO-VO** without access to an official implementation, **matching the reported ATE on TartanAir**.

Johns Hopkins University — *Sep 2025 — Feb 2026*
Research Assistant · Advisor: Prof. Anand Bhattad — Remote
- Developed a ViT-based latent encoder that disentangles illumination from scene content; on the **MIT Multi-Illumination** test set, reduced color-corrected relighting RMSE by **7.2%** (**0.222 to 0.206**) and improved SSIM from **0.571 to 0.600** over the published latent-intrinsics baseline.
- Designed a DiT-based generative pipeline using illumination as a prompt and reference scene content as control for high-quality, controllable image relighting.

The Hong Kong University of Science and Technology (Guangzhou) — *Nov 2023 — Jun 2024*
Research Assistant · Advisor: Prof. Haoang Li — Remote
- Developed an **animatable human Gaussian Splatting** pipeline using a canonical avatar representation and **SMPL-driven** deformation across per-frame body poses.
- Co-designed and implemented a correspondence-guided feature-consistency loss using **RoMA** matches and **DINO features** across viewpoints and body poses, complementing photometric supervision to enforce part-level appearance consistency under articulated motion.

SELECTED PROJECTS

Privacy-Aware V2V Cooperative Perception — *Jun 2025 — Jul 2025*
Supervised by Prof. Akarsh Prabhakara — Madison, WI
- Contributed to the development and evaluation of **SHARP**, a **privacy-aware V2V cooperative perception** framework that shares pointmap-based novel-view renderings instead of raw camera images.
- Evaluated **VGGT** on **OPV2V** across ego-only, SHARP-generated, and raw multi-agent inputs to assess 3D reconstruction under different data-sharing settings.
- Extended the **OPV2V/CARLA** simulation with **depth sensing** for point-cloud rescaling and downstream 3D recovery.

PUBLICATIONS

Privacy-Aware Sharing of Raw Spatial Sensor Data for Cooperative Perception — *HotMobile 2026*
*Bangya Liu, Chengpo Yan, **Chenghao Jiang**, Suman Banerjee, Akarsh Prabhakara*

SKILLS

Programming — *Python · CUDA · PyTorch · LaTeX*

Research — *3D reconstruction · SLAM · Gaussian Splatting · Diffusion Models · VLMs*

Last updated · Oct. 2026